ORSIDiff: Diffusion Model for Salient Object Detection in Optical Remote Sensing Images

Jinyu Han, Jing Sun, Fasheng Wang, Fuming Sun, and Haojie Li

***Abstract*— The unique imaging conditions of satellites introduce significant uncertainties in the structure and scale of ground objects, presenting a major challenge for optical remote sensing image salient object detection (ORSI-SOD). Current ORSI-SOD methods often fail to effectively differentiate between salient objects and subtle background variations, leading to suboptimal prediction outcomes. Furthermore, ORSI-SOD is a dense pixel prediction task, and existing approaches frequently depend on pixel-level probabilities, which can result in overconfident and inaccurate predictions. To address these challenges, we reformulate the ORSI-SOD task as a mask-generation problem by introducing a novel paradigm and propose a diffusion model-based method for ORSI-SOD, termed ORSIDiff. Central to our approach is the design of a powerful denoising network that enhances the model's refinement capabilities. This network leverages the strengths of both global and local modeling, improving the handling of salient object details and enabling a deeper understanding of the distinctions between salient objects and their surroundings. Additionally, we introduce a consistency assessment strategy (CAS) that aggregates multiple potential predictions during the denoising process, effectively mitigating the issue of overconfident point estimation. The extensive experimental results on two widely used ORSI-SOD datasets demonstrate that ORSIDiff achieves significant performance improvements over 20 state-of-the-art (SOTA) methods.**

***Index Terms*— Diffusion model (DM), optical remote sensing images (ORSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) seeks to identify and segment the most visually striking objects or regions within a scene by mimicking the human visual attention mechanism. The outcomes of this detection process are subsequently utilized in various downstream visual tasks, including quality assessment [1], image segmentation [2], [3], video compression [4], and object tracking [5]. The field of SOD initially concentrated on natural scene images (NSIs), producing a wealth of influential research that laid a robust groundwork for its application in the aforementioned visual domains.

With the increasing accessibility of optical remote sensing images (ORSIs) and the growing demand in fields such as agriculture, forestry, environmental science, and safety monitoring, ORSI-SOD has emerged as a prominent research focus. However, due to differences in optical equipment and imaging conditions, NSIs and ORSIs exhibit distinct characteristics. Most NSIs are typically captured from a horizontal perspective using devices like cameras or smartphones, primarily featuring everyday landscapes, urban environments, humans, and animals. In contrast, ORSIs are acquired from high altitudes through aerial sensors and satellites, predominantly containing objects such as airplanes, buildings, ships, rivers, roads, and other large-scale geographical features. These differences in imaging perspectives and content present unique challenges and opportunities for ORSI-SOD research.

Compared to NSIs, ORSIs often present more challenging scenes, including large objects, shadow interference, blurry boundaries, and complex backgrounds. Clearly, due to the unique characteristics of ORSIs, directly applying NSI-SOD models to ORSI-SOD tasks is unlikely to yield optimal detection results. Consequently, researchers have begun exploring SOD methods specifically tailored for ORSI scenarios. As a contribution to this effort, we propose a novel method that achieves superior detection results in challenging scenarios compared to state-of-the-art (SOTA) methods, such as WeightNet [6], RAGRNet [7], and PRNet [8], as shown in Fig. 1.

In general, ORSI-SOD can be formulated as a pixel-level dense prediction task. However, this paradigm, which relies heavily on pixel-level probabilities, is prone to overconfident point estimation, often resulting in incorrect predictions. Furthermore, ORSI-SOD frequently encounters complex scenes featuring numerous small objects, multiple objects, shadow occlusion, and insufficient lighting. Without carefully designed strategies, this discriminative paradigm can easily fail to distinguish subtle differences between salient objects and their surroundings, leading to suboptimal prediction results. Addressing these challenges requires innovative approaches to enhance the model's ability to handle such complexities effectively.

The rise of diffusion models has ushered in a new era of advancements across various computer vision tasks, such as image segmentation [9], [10] and object detection [11]. These models operate through a forward diffusion process and a reverse denoising process, both rooted in a fixed Markov chain.

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Jinyu Han, Jing Sun, Fasheng Wang, and Fuming Sun are with the School of Information and Communication Engineering, Dalian Minzu University, Dalian 116600, China (e-mail: hanjinyu163@gmail.com; jingsun@dlnu.edu.cn; wangfasheng@dlnu.edu.cn; sunfuming@dlnu.edu.cn).

Haojie Li is with the School of Computer Science and Engineering, Shandong University of Science and Technology, Qingdao 266590, China (e-mail: hjli@sdust.edu.cn).

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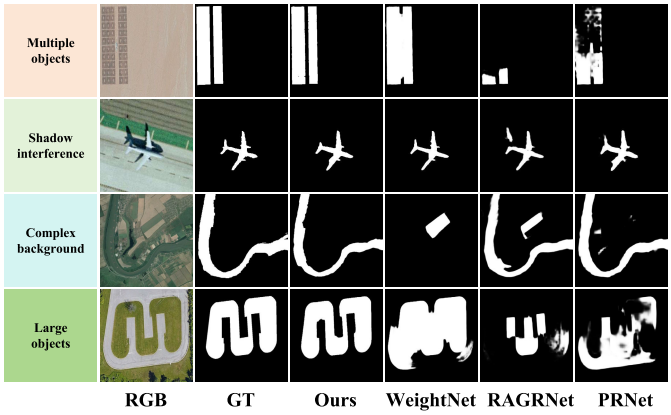


Fig. 1. Experimental results demonstrate the superiority of our proposed method compared to the other three methods (WeightNet [6], RAGRNet [7], and PRNet [8]) on complex scenarios, including multiple objects, shadow interference, complex background, and large objects.

During the forward process, noise is incrementally added to the target image, gradually transforming it to approximate a Gaussian distribution. Conversely, the reverse process learns to map the noise back to the original image. Inspired by the remarkable performance of diffusion models in numerous computer vision tasks, we integrate this approach into ORSI-SOD. To the best of authors' knowledge, the application of diffusion models to generate salient object images in the context of remote sensing remains an entirely unexplored frontier.

However, when directly applying some existing diffusion-based object detection methods (such as CamoDiff [12] and FocusDiff [13]) to the ORSI-SOD task, three key limitations arise: 1) the uniqueness of this task lies in its involvement with multiple complex scenes, including small objects, multiple objects, and shadow interference, which existing methods struggle to handle effectively; 2) their ability to distinguish between object and background boundaries is insufficient, often resulting in blurred target boundaries or missing details in the segmentation results; and 3) their inference efficiency is relatively low, which complicates their application in practical scenarios. These limitations highlight the need for a more tailored approach to address the specific challenges of ORSI-SOD.

To address the aforementioned challenges, we propose a novel ORSI-SOD framework, termed ORSIDiff, which reformulates the ORSI-SOD task as a conditional mask generation process based on the diffusion model. At its core, ORSIDiff performs diffusion on the mask during training and leverages the RGB image as a conditional cue during the denoising process to gradually guide the initial noise toward alignment with the ground truth. Given the unique challenges of ORSI-SOD, we introduce a denoising network based on a discriminative paradigm, designed to fully exploit the noisy mask. This network takes ORSI images as input and incorporates the noisy mask to predict previous states. Notably, the network no longer directly maps the input ORSI to the ground-truth mask; instead, it utilizes both the ORSI images and the noisy mask to generate a mapping of the noisy mask state at the previous time step. This approach plays a pivotal role in addressing the diverse and complex scenarios encountered in ORSI-SOD. To mitigate the efficiency concerns associated with diffusion models, the network employs a lightweight hybrid backbone network, SMT [14], which excels in both local and global modeling capabilities. Additionally, to reduce the interference caused by potential outliers during the reverse denoising process, we introduce a consistency assessment strategy (CAS). This strategy aims to filter out outliers, ensuring smoother object predictions and thereby minimizing the negative impact on detection performance. Together, these innovations enable ORSIDiff to effectively tackle the unique challenges of ORSI-SOD while maintaining computational efficiency.

Our main contributions are as follows.

1) We introduce a novel paradigm that reformulates ORSI-SOD as a conditional guided mask generation task based on the diffusion model. Building on this, we propose a groundbreaking ORSI-SOD method, named ORSIDiff, which leverages the diffusion model (DM) to significantly enhance the ability to address the unique challenges posed by ORSI-SOD. This approach not only improves the robustness and accuracy of SOD in complex remote sensing scenarios, but also sets a new direction for future research in this domain.
2) We propose a novel denoising network composed of an encoder and a decoder. The encoder employs a hybrid backbone to extract features, combining the strengths of global and local modeling. This dual capability enables a deeper understanding of the distinctions between salient objects and their surrounding environments, thereby enhancing the network's ability to process fine details of salient objects. The decoder is primarily designed with three key components: the feature grouping enhancement module (FGEM), the perceptual refinement module (PRM), and the refinement module (RM). These modules collaborate to refine and enhance feature representations, ensuring accurate and robust SOD.
3) We introduce a CAS to address the outliers generated at different time steps during the reverse denoising process. The CAS operates by filtering out inconsistent predictions, effectively reducing the interference caused by outliers. This strategy enhances the reliability and accuracy of the final detection results, ensuring more robust performance in complex ORSI-SOD scenarios.

II. Related Work

A. SOD for ORSIs

Recent years, with the advent of convolutional neural networks (CNNs) and Vision Transformer, the NSI-SOD domain has seen rapid advancements, leading to the development of various innovative techniques such as global context-aware aggregation [15], attention mechanisms [16], edge guidance [17], [18], and multiscale interaction [19], [20]. For example, Qin et al. [15] developed a nested U-structure for SOD for capturing more contextual cues with residual

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U-blocks, Li et al. [16] designed a multilevel attention cascaded feedback module to capture the cross-level complementary information, which can feedback features of high semantics to previous ones to refine them for better saliency prediction results, Zhao et al. [17] presented an edge guidance network for SOD with three steps to simultaneously model these two kinds of complementary information in a single network, and Liang et al. [18] introduced a multiscale attention-edge interactive refinement network to enable close interaction between attention and edges. Pang et al. [19] constructed an interaction component to aggregate similar information in an encoder and then used a self-interaction component to extract multiscale cues for the decoder. Hu et al. [20] designed a cross-modal fusion and progressive decoding network to achieve SOD tasks. Zhuge et al. [21] proposed a comprehensive cognitive network designed to learn robust integrity features, while Han et al. [22] introduced a perceptual localization and focus refinement network to explore consistency information across features at different levels and patterns. These advancements have significantly improved the capability to detect salient objects in complex natural scenes.

However, due to the inherent differences between NSIs ORSIs, directly applying NSI-SOD methods to ORSIs often yields suboptimal results. Consequently, researchers have increasingly focused on developing techniques specifically tailored for ORSI-SOD. This shift has led to the creation of a variety of innovative approaches, significantly enhancing the performance of SOD in remote sensing scenarios. For instance, Quan et al. [23] proposed a multilayer feature extraction framework utilizing a CNN backbone, integrated with a global semantic enhancement module and an uncertainty-aware RM to optimize feature embedding, thereby achieving more precise salient target localization and recognition. Zhao et al. [7] introduced a recursive adaptive graph reasoning network, specifically designed to capture the patterns of salient objects in complex and irregular remote sensing imagery. A multiscale multidimensional weighted network is proposed in [6], effectively addressing issues of scale adaptability and noise interference in direct multiscale feature fusion. Together, these advancements have propelled the ORSI-SOD field to new heights in terms of both performance and robustness.

Additionally, the methods mentioned above often come with a high number of parameters and significant computational complexity, which restricts their deployment on resource-constrained mobile devices and creates a gap between scientific research and practical applications. To address this, a few researchers have started developing lightweight ORSI-SOD methods, such as [7] and [24]. While these lightweight approaches focus on reducing parameter counts and computational costs, they typically employ shallower network architectures, which limits their feature representation capabilities compared to more complex models. As a result, these methods struggle to effectively tackle the inherent challenges of ORSI-SOD, often exhibiting a noticeable performance gap when compared to their more computationally intensive counterparts. This highlights the need for innovative solutions that balance efficiency and effectiveness in ORSI-SOD tasks.

B. Diffusion Models

In recent years, DM [25], [26] has emerged as a powerful class of deep generative models, achieving SOTA performance across various domains, including images [27], [28], videos [29], [30], [31], and audio [32].

A DM consists of two primary processes: a forward process and a reverse process. The forward process gradually adds noise to the original data until it becomes pure Gaussian noise, while the reverse process incrementally removes noise to generate samples that align with the object distribution. Leveraging their exceptional generative capabilities, researchers have begun integrating diffusion models into dense visual perception tasks such as image segmentation [10], [33], depth estimation [34], [35], and object detection [12], [36]. These efforts demonstrate the versatility and potential of diffusion models in advancing complex visual tasks.

For instance, Baranchuk et al. [9] extended the DM to image segmentation tasks, demonstrating its potential in this domain. Wolleb et al. [37] further explored its application in medical image segmentation, showcasing its effectiveness in specialized fields. Mascaro et al. [38] applied the DM to 3-D semantic segmentation, highlighting its versatility in handling complex spatial data. Ke et al. [39] investigated the potential of diffusion models in depth estimation, achieving promising results. Chen et al. [12] successfully adapted the DM for camouflaged object detection, addressing the unique challenges of identifying objects that blend into their surroundings. Additionally, Li et al. [36] proposed a novel framework based on diffusion models to tackle the imbalance in the distribution of visual and language subspaces in open lexical object detection, further broadening the scope of DM applications. These studies collectively underscore the adaptability and effectiveness of diffusion models across a wide range of visual perception tasks.

Given the remarkable success of the DM in the areas mentioned above, we introduce it to the ORSI-SOD task. However, applying the existing DM directly to ORSI-SOD presents several challenges. First, due to the unique complexities of ORSI, such as large-scale scenes, shadow interference, and complex backgrounds, the image features generated by traditional diffusion methods may struggle to restore salient masks or achieve sufficient mask refinement accurately. Second, DM typically involves a large number of parameters and high computational complexity, which limits its practicality in resource-intensive visual tasks like object detection. Finally, the inherent iterative generation mechanism of diffusion models, which operates over multiple time steps, results in low inference efficiency. This further complicates their deployment in real-world applications.

To address these challenges, we propose a tailored network designed to enhance the applicability and efficiency of diffusion models for ORSI-SOD, ensuring they can effectively handle the unique demands of remote sensing imagery. This network achieves a balance between maintaining high detection accuracy and significantly improving computational efficiency, demonstrating the strong potential of diffusion models in tackling the complexities of remote sensing scenes.

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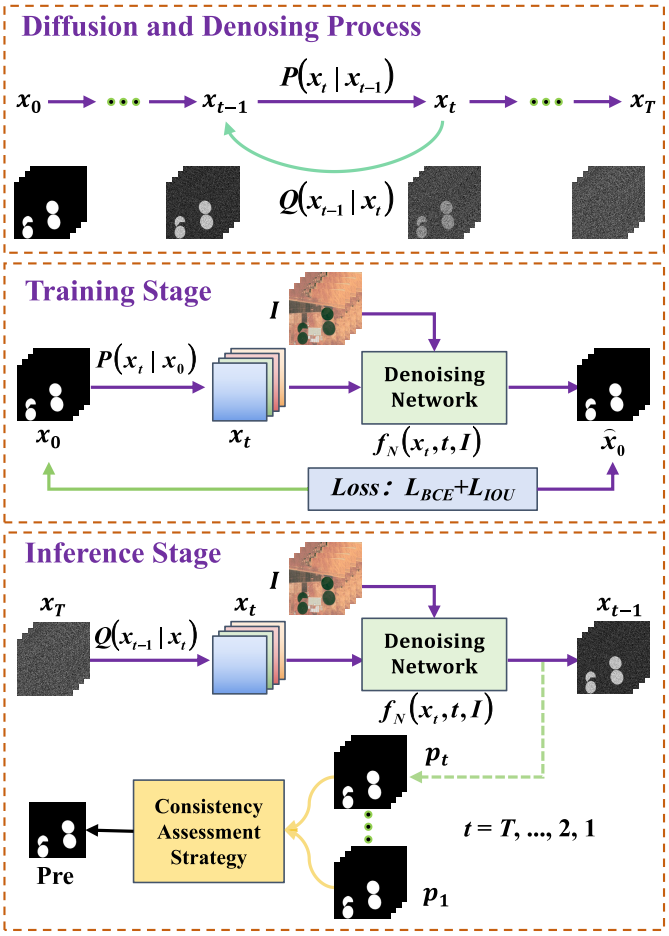


Fig. 2. Principle of ORSI-SOD network based on diffusion model, which consists of training and inference stages. Conventional diffusion models employ an unconditional "noise-addition-and-removal" paradigm for image generation. We transform ORSI-SOD into a conditional mask generation task: at each denoising step of the inference stage, the model integrates: 1) hierarchical prior features from the input image I and 2) the current noisy mask state x_t through a bimodal fusion mechanism. Through iterative optimization, the proposed model progressively enhances the detailed representation of salient objects, ultimately outputting precise salient region detection results via a CAS.

By addressing the unique challenges of ORSI-SOD, our approach demonstrates how diffusion models can be effectively adapted for practical applications in remote sensing, paving the way for further advancements in this field.

III. Proposed Method

A. Network Overview

In this section, we describe the framework of the diffusion-based ORSI-SOD method in detail. The principle of the method is shown in Fig. 2. During the forward diffusion process, Gaussian noise is gradually introduced to an image x_0, resulting in a series of noisy versions x_t over time steps $t \in [1, T]$. In the reverse denoising process, the noisy mask x_t is iteratively refined to approximate the original image x_0. Finally, a CAS is employed to aggregate all intermediate estimates $\{p_t\}_{t=1}^{T}$ and produce the final predicted result **Pre**. This approach leverages the diffusion model's ability to handle complex data distributions, making it well-suited for addressing the unique challenges of SOD in ORSIs.

B. Diffusion Model

1) Forward Diffusion Process: The process can be considered as a Markov chain that gradually adds noise to the data through T time steps. At each time step t, the noisy mask x_t can be expressed as

$$P(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{\alpha_t}x_{t-1}, \sqrt{1-\alpha_t}\mathcal{I}\right) \tag{1}$$

where $\mathcal{N}$ denotes the normal distribution with a specified mean and variance, $\mathcal{I}$ denotes the standard normal distribution, and $\alpha_t \in (0, 1)$ is a predefined noise schedule that controls the variance of noise added at time step t. More specifically, given the original input x_0, the marginal distribution of x_t can be described as

$$P(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathcal{I}\right) \tag{2}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$.

2) Reverse Denoising Process: Starting from x_T, the denoising network $f_N(\cdot)$ is employed to iteratively remove noise from the noisy mask x_t, recovering the previous noisy mask x_{t-1}. Given the noisy mask x_t, the probability distribution of x_{t-1} can be expressed as

$$\begin{cases} Q(x_{t-1}|x_t) = \mathcal{N}\left(x_{t-1}; \mu_N(x_t, t), \sigma_t^2\mathcal{I}\right) \\ \sigma_t^2 = \dfrac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}(1-\alpha_t) \end{cases} \tag{3}$$

where α_t directly determines the variance σ_t^2, and $\mu_N(\cdot)$ is formulated as

$$\mu_N(x_t, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\hat{x}_0 \tag{4}$$

where $\hat{x}_0$ is the output of the denoising network during the training stage.

C. Denoising Network

To predict and remove the noise introduced during the forward diffusion process, we design a denoising network $f_N(\cdot)$, which comprises a feature encoder and a decoder. Unlike traditional segmentation approaches, our proposed model leverages the input image I as a conditional cue to guide the DM in generating controllable predictions. As illustrated in Fig. 3, the image I and the noisy mask x_t are jointly fed into the denoising network to estimate the ground truth. Further details of the network architecture are discussed in the following paragraphs.

1) Encoder: Given the unique challenges presented by ORSI, including variable object scales, uncertain numbers of objects, the potential absence of salient objects, and cluttered backgrounds, it is crucial to fully harness the guiding role of the input image. To enable the network to discern the subtle distinctions between salient objects and complex backgrounds, the feature encoder must be equipped with both global and local modeling capabilities. This dual capability ensures that the network can effectively address these challenges and accurately identify salient regions amidst diverse and intricate scenarios. Traditional diffusion models typically employ U-Net or Transformer architectures to progressively drive

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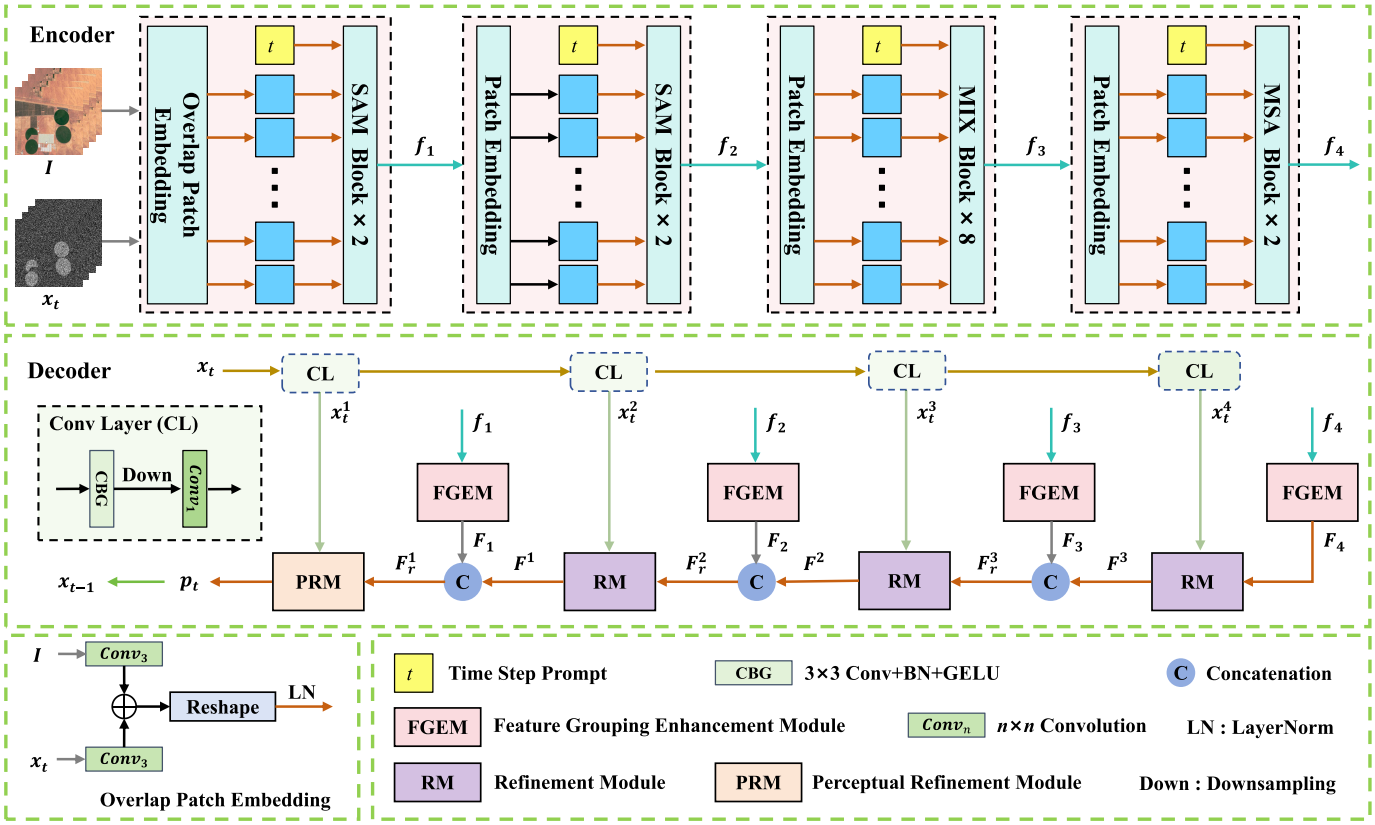


Fig. 3. Framework of the proposed denoising network. The FGEM, RM, and PRM are illustrated in Figs. 4–6, respectively. The denoising network (i.e., the conditional mask generation framework) employs an encoder–decoder architecture. The encoder utilizes SMT-a hybrid CNN-Transformer structure that achieves optimal balance between parameter efficiency and feature encoding capability. The decoder incorporates three specialized modules: 1) feature grouping enhancement module; 2) RM; and 3) PRM. They work together to address the challenges faced by ORSI-SOD.

the denoising process. However, while CNNs excel in local modeling and Transformers perform well in global modeling, both architectures require more parameters and higher computational costs, which can hinder the efficiency of diffusion models. Therefore, striking a balance between performance and computational feasibility is crucial for denoising tasks.

Based on the aforementioned analysis, we select a lightweight hybrid backbone, SMT [14], as the encoder for feature extraction. This choice is driven by its dual capability in local and global modeling while maintaining computational efficiency. The SMT backbone comprises four stages: the first two stages utilize the scale-aware modulation (SAM) module, the third stage incorporates the multihead mixed convolution (MIX) module, and the fourth stage employs the multihead self-attention (MSA) module. In our framework, the encoder's role is to integrate the input image and noisy mask, generating a more robust, refined, and feature-rich representation. This approach fully leverages the guiding role of the input image, enhancing the model's ability to discern subtle differences between salient objects and complex backgrounds.

Specifically, we adapt the structure of SMT to achieve denoising, as illustrated in Fig. 3. In this modified architecture, overlapping patch embedding replaces the standard patch embedding in the first layer of SMT. When the image I and the noisy mask x_t are fed into the encoder, they are first mapped to the same channel dimension using convolution operations. Subsequently, an elementwise addition operation is applied to combine the two inputs. This design ensures that the original Transformer structure and pretrained parameters remain intact, preserving the model's efficiency and effectiveness. This process can be described as

$$B_i = \begin{cases} \mathrm{OPE}(I, x_t), & i = 1 \\ \mathrm{PE}(f_{i-1}), & i = 2, 3, 4 \end{cases} \tag{5}$$

where $\mathrm{PE}(\cdot)$ refers to the patch embedding, and $\mathrm{OPE}(\cdot)$ is the overlap patch embedding, which is given as

$$\mathrm{OPE}(I, x_t) = \mathrm{LN}(\mathrm{Reshape}(\mathrm{Conv}_3(I) + \mathrm{Conv}_3(x_t))) \tag{6}$$

where "+" means the elementwise addition, $\mathrm{Conv}_3(\cdot)$ is the convolution with kernel 3×3, $\mathrm{Reshape}(\cdot)$ means that the feature map is converted to tokens, and $\mathrm{LN}(\cdot)$ represents the LayerNorm.

In the diffusion model, noise levels and recovery difficulty vary significantly across time steps. Specifically, in the early stages, the noise level is relatively low, preserving more significant information and making denoising tasks easier. However, as the time steps progresses, the noise level gradually increases, and the most salient information becomes almost entirely obscured, significantly increasing the difficulty of recovery.

To address this issue, we introduce time-step embedding, which is designed to capture the dynamic characteristics of different time steps. Time-step embedding provides the model with essential information about the current stage of

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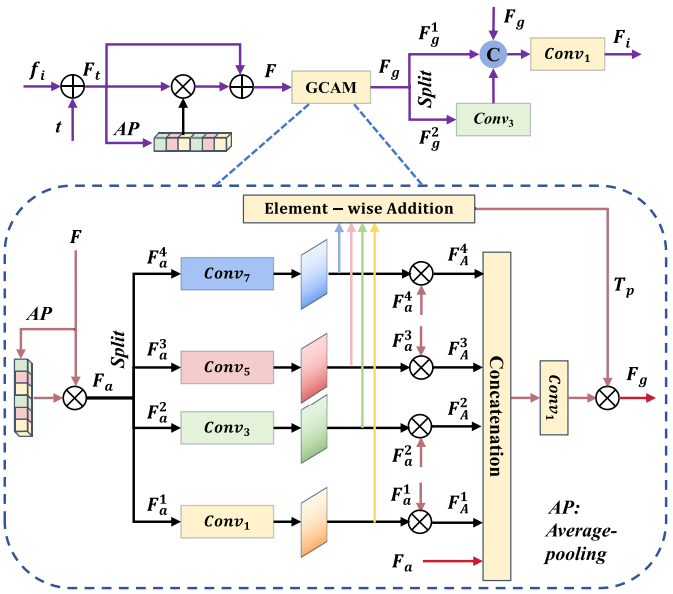


Fig. 4. Illustration of the FGEM. It focuses on identifying and extracting dominant intrinsic consistency information within the features.

the diffusion process, enabling it to learn appropriate feature representations at each time step. Through this mechanism, the model enhances its adaptability to noise characteristics and makes precise denoising decisions based on the specific state of each time step. Notably, during the later time steps, time-step embedding helps the model focus on more complex patterns and structures, thereby facilitating a more effective recovery of salient information.

Inspired by the task domain prompts in [40], we incorporate the time step t to assist the network in intentionally performing denoising at different stages of the diffusion process. This process can be described as

$$f_i = \begin{cases} \text{SAM}(\text{Cat}(t, B_i)), & i = 1, 2 \\ \text{MIX}(\text{Cat}(t, B_i)), & i = 3 \\ \text{MSA}(\text{Cat}(t, B_i)), & i = 4 \end{cases} \tag{7}$$

where $\text{Cat}(\cdot)$ denotes the concatenation operation, $\text{SAM}(\cdot)$ refers to the SAM module, $\text{MIX}(\cdot)$ means the MIX module, and $\text{MSA}(\cdot)$ refers to the MSA module, as shown in Fig. 3.

2) FGEM: Due to the challenges in ORSI-SOD, such as uncertainty in the number and scale of objects, the absence of salient objects, blurred boundaries caused by shadows, and complex backgrounds, accurately capturing salient objects can be particularly difficult. To address these issues, as illustrated in Fig. 4, we propose an FGEM. This module is designed to group and enhance the feature embeddings from the encoder, extracting dominant intrinsic consistency information from the features to support subsequent decoding processes. Additionally, we introduce the time step t as a prompt to further refine the denoising process across different time steps, enhancing the model's ability to adapt to varying noise levels and recovery requirements.

Specifically, we first transform the time step t from a token into a feature map with the same dimension as the input feature f_i. We then perform an elementwise addition operation between them to obtain feature F_t, effectively embedding the denoising time step t into the feature representation. This process not only enhances the model's perception of time step information, but also facilitates subsequent feature optimization. Next, we generate an attention map by applying an average-pooling operation on the feature F_t along the channel dimension. This attention map is used to refine the feature F_t, and an elementwise addition operation is performed to produce the feature F. It is fed into the grouped convolutional attention mechanism (GCAM), which aims to further extract intrinsic consistent information from F and enhance its dominant features, generating the feature F_g. Subsequently, F_g is split into two parts using a splitting operation. One part undergoes a 3×3 convolution operation to adaptively preserve or exclude certain pixels. The results of this operation are then concatenated with the original feature F_g to prevent the loss of useful information. Finally, a 1×1 convolution operation is applied to adjust the number of channels, producing the output feature F_i.

This process can be described as

$$\begin{cases} F = \text{AP}(F_t) \times F_t + F_t \\ F_g = \text{GCAM}(F) \\ F_g^1, F_g^2 = \text{Split}[F_g] \\ F_i = \text{Conv}_1\left(\text{Cat}\left[F_g, \text{Conv}_3\left(F_g^2\right), F_g^1\right]\right) \end{cases} \tag{8}$$

where $\text{AP}(\cdot)$ represents the average-pooling operation, $\text{GCAM}(\cdot)$ refers to the GCAM, "$\times$" is the elementwise multiplication, $\text{Split}[\cdot]$ refers to the splitting operation, $\text{Conv}_n(\cdot)$ denotes the $n \times n$ convolution, and $\text{Cat}[\cdot]$ represents the concatenation operation.

For the GCAM illustrated in Fig. 4, we begin by dividing the features into multiple groups of subfeatures. We then leverage the correlations between different channels and contextual information at various scales to extract salient information from these subfeatures. Finally, we enhance the features by aggregating the subfeatures, ensuring a more robust representation of the dominant information. This process can be described as

$$\begin{cases} F_a = \text{AP}(F) \times F \\ F_a^1, F_a^2, F_a^3, F_a^4 = \text{Split}[F_a] \\ F_A^i = F_a^i \times \text{Conv}_n\left(F_a^i\right) \\ \qquad i, n = (1, 1), (2, 3), (3, 5), (4, 7) \\ T_p = \text{Conv}_1\left(F_a^1\right) + \text{Conv}_3\left(F_a^2\right) + \text{Conv}_5\left(F_a^3\right) \\ \qquad + \text{Conv}_7\left(F_a^4\right) \\ F_g = \text{Conv}_1\left(\text{Cat}\left[F_A^1, F_A^2, F_A^3, F_A^4\right]\right) \times T_p. \end{cases} \tag{9}$$

D. Decoder

Unlike previous diffusion models, we integrate the noisy mask into the decoding process to enhance the refinement of the noisy mask. This integration enables the network to gain a more comprehensive understanding of features during decoding, thereby improving its ability to accurately predict

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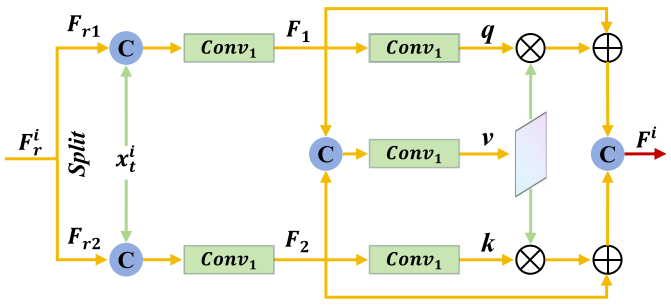


Fig. 5. Illustration of the RM. It is designed to enhance and clarify blurred details by leveraging the diverse feature representations of subfeatures.

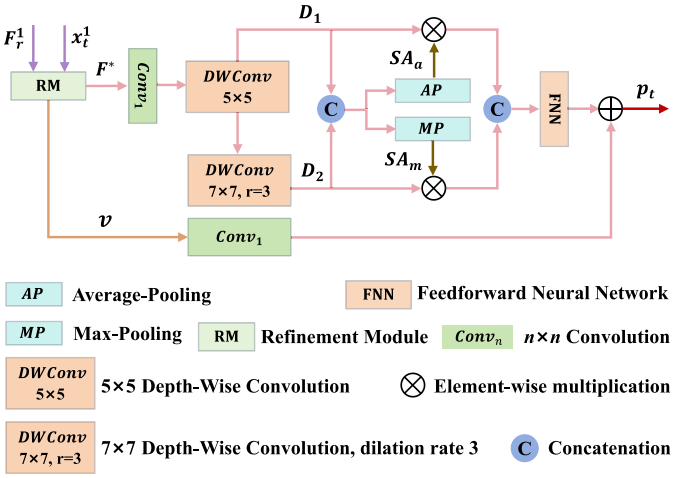


Fig. 6. Illustration of the PRM. It is devised to model both global and local perceptions, enabling more precise refinement of local details while simultaneously enhancing the overall salient information.

and remove noise. As shown in Fig. 3, the decoder utilizes the convolutional layer (CL) to adjust the size and channel number of the noisy mask x_t, ensuring consistency with the four-level features F_i (where $i = 1, 2, 3$, and 4) generated by the FGEM. The overall process can be described as follows:

$$x_t^i = \begin{cases} \mathrm{CL}(x_t), & i = 1 \\ \mathrm{CL}\left(x_t^{i-1}\right), & i = 2, 3, 4 \end{cases} \tag{10}$$

where $\mathrm{CL}(\cdot)$ represents the convolutional layer.

Afterward, we execute top–down decoding, gradually fusing lower level features starting from the highest-level feature F_4. Notably, we do not rely on stacked CLs or blocks from existing backbone networks to guide the decoding process. Instead, we design a simple RM and PRM to refine the features, with a particular focus on enhancing the details of locally ambiguous features. Compared to previous methods, our approach significantly reduces the number of parameters and computational load, addressing the issue of low efficiency commonly associated with diffusion models. The process is described as

$$\begin{cases} F^3 = \mathrm{RM}\left(x_t^4, F_4\right) \\ F^i = \mathrm{RM}\left(x_t^{i+1}, F_r^{i+1}\right), \quad i = 1, 2 \\ F_r^i = \mathrm{Cat}[F^i, F_i], \quad i = 1, 2, 3 \\ p_t = \mathrm{PRM}\left(x_t^1, F_r^1\right) \end{cases} \tag{11}$$

where $\mathrm{RM}(\cdot)$ is the refinement module, $\mathrm{PRM}(\cdot)$ represents the perceptual refinement module, and $\mathrm{Cat}[\cdot]$ denotes the concatenation operation.

1) RM: The RM structure is shown in Fig. 5. RM aims to embed the noisy mask x_t into the subfeatures F_{r1} and F_{r2} of the input feature to refine the blurred details based on the diverse feature representation of the subfeatures. Specifically, for the convenience of explanation, we refer to the input feature as F_r^i ($i = 1, 2$, and 3). First, the F_r^i is split into two subfeatures F_{r1} and F_{r2} along the channel dimension, and the noisy mask x_t^i is introduced into the two subfeatures. Subsequently, F_{r1} and F_{r2} use convolution operations to further fuse the noisy mask and adjust the number of channels, respectively, named F_1 and F_2. Inspired by the self-attention mechanism, we concatenate F_1 and F_2 and use convolution operations to change the number of channels to 1, thereby obtaining the attention map v. It should be explained that using convolution operations to get v can enhance the correlation between the channels of the two subsfeatures. Subsequently, we perform convolution operations on F_1 and F_2 to learn new feature representations and thus obtain q and k, respectively. Then, we use v to align q and k, respectively. Next, we perform elementwise addition operations on the aligned q and k with F_1 and F_2, respectively. Finally, we concatenate them along the channel dimension to obtain the final output F^i. The process is described as

$$\begin{cases} F_{r1}, F_{r2} = \mathrm{Split}\left[F_r^i\right] \\ F_1 = \mathrm{Conv}_1\left(\mathrm{Cat}\left[F_{r1}, x_t^i\right]\right) \\ F_2 = \mathrm{Conv}_1\left(\mathrm{Cat}\left[F_{r2}, x_t^i\right]\right) \\ v = \mathrm{Conv}_1(\mathrm{Cat}[F_1, F_2]) \\ q = \mathrm{Conv}_1(F_1) \\ k = \mathrm{Conv}_1(F_2) \\ F^i = \mathrm{Cat}\left[(q \times v + F_1), (k \times v + F_2)\right]. \end{cases} \tag{12}$$

2) PRM: The structure of PRM is illustrated in Fig. 6. PRM is to model both global and local perceptions, enabling more precise refinement of local details and enhancement of overall salient information. Specifically, F_r^1 and the noisy mask x_t^1 are first input into the RM to undergo feature fusion and refinement, yielding feature representation F^* and the attention map v. Subsequently, two large convolution kernels with different scales are adopted to aggregate contextual information, resulting in features D_1 and D_2. Inspired by the integration of max pooling and average pooling in convolutional block attention module (CBAM), D_1 and D_2 are concatenated and processed in the spatial dimension using average pooling and max pooling to generate spatial attention maps SA_a and SA_m, respectively. These attention maps are then used to calibrate D_1 and D_2 independently. The calibrated features are concatenated again and fed into a feedforward neural network (FNN) to improve their representation capacity. Finally, v is adaptively adjusted via convolution operation and further utilized to enhance salient feature information, generating the output p_t. The detailed process is described

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as follows:

$$\begin{cases} F^* = \mathrm{RM}\left(x_t^1, F_r^1\right) \\ D_1 = \mathrm{DWConv}_5\left(\mathrm{Conv}_1\left(F^*\right)\right) \\ D_2 = \mathrm{DWConv}_7(D_1) \\ \mathrm{SA}_a = \mathrm{AP}(\mathrm{Cat}[D_1, D_2]) \\ \mathrm{SA}_m = \mathrm{MP}(\mathrm{Cat}[D_1, D_2]) \\ p_t = \mathrm{FNN}(\mathrm{Cat}[\mathrm{SA}_a \times D_1, \mathrm{SA}_m \times D_2]) \\ \qquad + \mathrm{Conv}_1(v) \end{cases} \tag{13}$$

where $\mathrm{DWConv}_n(\cdot)$ represents the $n \times n$ depthwise convolution, $\mathrm{FNN}(\cdot)$ denotes the FNN, and $\mathrm{MP}(\cdot)$ is the max-pooling operation.

3) CAS: The DM can produce diverse results during the inference stage. Inspired by the annotation procedure of saliency detection [41] and consensus time ensemble strategy [12], we introduce a CAS to integrate the diverse results $p_t, (t = 1, 2, 3, \ldots, T)$. Specifically, for multiple predictions p_t $(t = 1, 2, \ldots, T)$, we first implement normalization and binary operations on them through adaptive thresholding to get the corresponding binary masks p_t^b $(t = 1, 2, \ldots, T)$. Next, to generate a candidate mask, we use these predictions p_t^b to vote for the position of each point. In other words, we add up all T binary masks to obtain a mean result, which serves as the probability of candidate points. Finally, we set an experimental threshold and binarized these candidate points to obtain a desired mask. A visual explanation regarding the confidence level setting will be provided in Section IV.

The process is described as

$$\begin{cases} p_t^b = \mathrm{Round}(\mathrm{Norm}(p_t)), \quad t = 1, 2, \ldots, T \\ M_a = \dfrac{1}{T} \cdot \displaystyle\sum_{t=1}^{T} p_t^b \\ M_d = \mathrm{Round}(M_a + 0.1) \\ \mathbf{Pre} = M_d \times p_1 \end{cases} \tag{14}$$

where **Pre** refers to the output of the CAS, and $\mathrm{Round}(\cdot)$ means the binary operation. Based on the experimental results, M_d takes a value of 1 if $M_a \geq 0.4$, and 0 otherwise.

4) Loss Function: We utilize a hybrid loss function consisting of BCE loss [42] and IoU loss [43] to supervise our model during the training stage. The BCE loss focuses on pixel-level details, while the IoU loss focuses on foreground–background regions. The overall loss function is given as follows:

$$L = L_{\mathrm{BCE}}(\hat{x}_0, x_0) + L_{\mathrm{IoU}}(\hat{x}_0, x_0) \tag{15}$$

where $\hat{x}_0$ is the prediction result of the ground-truth map x_0, L_{BCE} represents the BCE loss, and L_{IoU} refers to the IoU loss.

IV. Experiments

A. Experimental Setup

1) Datasets: We evaluate our proposed ORSIDiff on two widely used datasets: EORSSD [44] and ORSSD [45]. Specifically, the EORSSD dataset consists of 2000 image pairs, with 1400 pairs used for training and 600 pairs for testing; the ORSSD dataset contains 800 image pairs, with 600 pairs allocated for training and 200 pairs for testing. Following the training strategy outlined in [46], we train ORSIDiff on the training set of each dataset and evaluate its performance on the corresponding test set.

2) Network Implementation: Our model is trained on a single NVIDIA RTX 3090 GPU. During training and testing, we resize the input images to 384×384 pixels. For the training procedure, we apply various data augmentation techniques, including random flipping, normalization, random cropping, and rotation, to enhance the robustness and generalization of the model. The training process utilizes the Adam optimizer [47], with a batch size of 8 and a learning rate of 1×10^{-4}. These settings ensure a stable and efficient training, enabling ORSIDiff to achieve optimal performance on the ORSI-SOD task. We adopt the SMT [14] backbone network, initialized with pretrained weights, while the remaining components are initialized using PyTorch's default settings. In addition, we empirically set the number of time steps $T = 10$ for the sampling process. This choice balances computational efficiency and the quality of the detected results, ensuring that the DM effectively refines the salient object predictions while maintaining reasonable inference times.

3) Evaluation Metrics: We adopt several widely used indicators to comprehensively evaluate the performance of our model, including F-measure (F_β, $\beta^2 = 0.3$) [48], S-measure (S_α, $\alpha = 0.5$) [49], E-measure (E_ξ) [50], mean absolute error ($\mathcal{M}$) [51], precision–recall (P–R) curve, and F-measure curve. These metrics collectively provide a robust evaluation framework, ensuring a thorough assessment of the model's accuracy, robustness, and generalization capabilities.

B. Comparison With SOTA

We conduct experiments on the EORSSD and ORSSD datasets to compare our proposed ORSIDiff with SOTA methods, including both NSI-SOD and ORSI-SOD approaches, as well as diffusion-based image segmentation methods. NSI-SOD method includes ICON [21]. ORSI-SOD methods include ERPNet [52] ACCoNet [53], AESINet [54], GeleNet [46], MCP-Net [55], ASNet [56], SFANet [23], SDPCNet [57], PRNet [8], RAGRNet [7], HFCNet [58], ADSTNet [59], LSHNet [60], WeightNet [6], SeaNet [61], ISAANet [24], and MEANet [62]. Diffusion models for image segmentation methods include CamoDiff [12] and FocusDiff [13]. The saliency maps of the above methods are sourced from the authors or generated by running publicly available source code.

1) Quantitative Evaluation: In Table I, we report the quantitative comparison results of ORSIDiff and 20 competing methods on the EORSSD and ORSSD datasets. As seen, ORSIDiff, with 16.98M parameters and 15.25G FLOPs, achieves nearly the best performance across all evaluation metrics. This demonstrates the efficiency and effectiveness of ORSIDiff in handling ORSI-SOD tasks, outperforming existing methods while maintaining a relatively lightweight architecture. These results highlight the robustness and superiority of ORSIDiff in detecting salient objects in complex remote sensing scenes.

Specifically, on the EORSSD dataset, compared to the HFCNet method, ORSIDiff achieves improvements of 0.33%,

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TABLE I

QUANTITATIVE COMPARISONS WITH 20 SOTA METHODS. RED AND BLUE DENOTE THE BEST AND THE SECOND-BEST RESULTS, RESPECTIVELY. "-" INDICATES ITS CODES OR RESULTS ARE NOT AVAILABLE

Methods	Params (M)	FLOPs (G)	FPS (GPU)	EORSSD								ORSSD							
				$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
ORSI-SOD Method and NSI-SOD Method																			
$ICON_{23}$	33.1	2.97	25	0.9185	0.8622	0.8371	0.8065	0.9687	0.9619	0.9497	0.0073	0.9256	0.8939	0.8671	0.8444	0.9704	0.9637	0.9554	0.0116
$ERPNet_{23}$	56.48	215.4	17	0.9210	0.8632	0.8304	0.7554	0.9603	0.9401	0.9228	0.0089	0.9254	0.8974	0.8745	0.8356	0.9710	0.9566	0.9520	0.0135
$ACCoNet_{23}$	102.55	348.8	10	0.9290	0.8837	0.8552	0.7969	0.9727	0.9653	0.9450	0.0074	0.9437	0.9149	0.8971	0.8806	0.9796	0.9754	0.9721	0.0088
$AESINet_{23}$	41.05	55.0	25	0.9358	0.8838	0.8524	0.7923	0.9751	0.9636	0.9462	0.0079	0.9460	0.9183	0.8986	0.8666	0.9828	0.9747	0.9707	0.0086
$GeleNet_{23}$	25.45	11.7	30	0.9376	0.8923	0.8781	0.8641	0.9828	0.9766	0.9750	0.0064	0.9469	0.9254	0.9128	0.9035	0.9860	0.9815	0.9786	0.0079
$MCP\text{-}Net_{24}$	25.1	86.6	-	0.9373	0.8868	0.8620	0.8077	0.9765	0.9657	0.9515	0.0070	0.9433	0.9135	0.8979	0.8726	0.9807	0.9736	0.9724	0.0090
$ASNet_{24}$	65.43	20.1	-	0.9345	0.8959	0.8770	0.8672	0.9783	0.9745	0.9748	0.0055	0.9441	0.9172	0.9072	0.8986	0.9803	0.9764	0.9795	0.0081
$SFANet_{24}$	25.1	7.7	-	0.9349	0.8833	0.8680	0.8492	0.9769	0.9726	0.9669	0.0058	0.9453	0.9192	0.9063	0.8984	0.9830	0.9789	0.9765	0.0077
$SDPCNet_{24}$	35.1	42.5	-	0.9391	0.9191	0.9044	0.8814	0.9821	0.9776	0.9713	0.0059	0.9425	0.9372	0.9239	0.9172	0.9836	0.9799	0.9770	0.0075
$PRNet_{24}$	20.8	8.47	21	0.9296	0.8707	0.8510	0.8282	0.9809	0.9732	0.9625	0.0053	0.9468	0.9153	0.9013	0.8915	0.9866	0.9824	0.9779	0.0068
$RAGRNet_{24}$	35.6	17.8	18	0.9361	0.8852	0.8770	0.8644	0.9788	0.9737	0.9749	0.0057	0.9507	0.9242	0.9157	0.9104	0.9861	0.9818	0.9800	0.0066
$HFCNet_{24}$	148.99	160.71	4	0.9449	0.8951	0.8646	0.8051	0.9842	0.9720	0.9552	0.0049	0.9543	0.9269	0.9049	0.8710	0.9884	0.9798	0.9748	0.0070
$ADSTNet_{24}$	62.09	27.72	-	0.9373	0.8868	0.8620	0.8077	0.9765	0.9657	0.9515	0.0070	0.9433	0.9135	0.8979	0.8726	0.9807	0.9736	0.9724	0.0090
$LSHNet_{24}$	-	-	-	0.9370	0.8844	0.8643	0.8261	0.9761	0.9692	0.9619	0.0064	0.9491	0.9200	0.9054	0.8848	0.9824	0.9764	0.9758	0.0075
$WeightNet_{24}$	103.65	143.9	-	0.9427	0.8955	0.8800	0.8437	0.9808	0.9760	0.9674	0.0053	0.9536	0.9281	0.9159	0.9043	0.9863	0.9826	0.9801	0.0068
$SeaNet_{23}$	2.76	1.7	75	0.9208	0.8649	0.8519	0.8304	0.9710	0.9651	0.9602	0.0073	0.9260	0.8942	0.8772	0.8625	0.9767	0.9722	0.9670	0.0105
$ISAANet_{24}$	3.59	4.34	108	0.9351	0.8869	0.8748	0.8643	0.9761	0.9717	0.9734	0.0063	0.9429	0.9166	0.9034	0.8982	0.9836	0.9787	0.9777	0.0077
$MEANet_{24}$	3.27	9.62	33	0.9282	0.8766	0.8678	0.8557	0.9708	0.9658	0.9663	0.0070	0.9340	0.9033	0.8934	0.8860	0.9768	0.9730	0.9715	0.0098
Diffusion-based Methods																			
$CamoDiff_{24}$	71.74	60.29	0.3	0.9298	0.8809	0.8610	0.8715	0.9685	0.9484	0.9673	0.0075	0.9410	0.9116	0.8913	0.8908	0.9779	0.9651	0.9706	0.0096
$FocusDiff_{24}$	166.18	3810.09	0.1	0.9184	0.8536	0.8429	0.8405	0.9591	0.9507	0.9450	0.0091	0.9352	0.9195	0.8896	0.8605	0.9707	0.9550	0.9652	0.0101
Ours	16.98	15.25	8	0.9482	0.9005	0.8885	0.8865	0.9850	0.9812	0.9831	0.0042	0.9580	0.9318	0.9264	0.9281	0.9893	0.9856	0.9858	0.0051

0.54%, 2.39%, 8.14%, 0.08%, 0.92%, 2.79%, and 0.07% in S_α, $F_\beta^{\max}$, F_β^{mean}, F_β^{adp}, $E_\xi^{\max}$, E_ξ^{mean}, E_ξ^{adp}, and $\mathcal{M}$, respectively. On the ORSSD dataset, ORSIDiff improves these indicators by 0.37%, 0.49%, 2.15%, 5.71%, 0.09%, 0.58%, 1.1%, and 0.19% over HFCNet, respectively. These results highlight that ORSIDiff exhibits significantly stronger saliency detection capabilities in complex remote sensing scenes, outperforming HFCNet and other competing methods across multiple evaluation metrics. This demonstrates the effectiveness of ORSIDiff in addressing the unique challenges of ORSI-SOD tasks.

In addition, we present the P–R curves and F-measure curves for both datasets in Fig. 7. On both the ORSSD and EORSSD datasets, ORSIDiff demonstrates the best performance on the P–R curve, surpassing all SOTA methods. Furthermore, the F-measure curve shows that ORSIDiff consistently maintains its advantage and performs robustly across various thresholds. These results underscore the superior detection accuracy and robustness of ORSIDiff, highlighting its ability to handle diverse and challenging scenarios in ORSI-SOD tasks.

2) Qualitative Evaluation: A visual comparison with SOTA methods is shown in Fig. 8. It can be observed that the proposed ORSIDiff generates more accurate saliency predictions across various challenging scenarios, showcasing its competitive edge over other SOTA models. Specifically, ORSIDiff excels in detecting objects of different sizes (first row), identifying large objects (second row), handling scenes without salient objects (third row), detecting multiple objects (fourth row), recognizing small objects (fifth row), overcoming objects in shadow interference (sixth row), navigating objects in complex background (seventh row), and resolving objects with blurred boundaries (eighth row).

Furthermore, we observe that other methods struggle in most ORSI-SOD challenging scenarios, often resulting in missed salient objects and erroneous detections. Notably, ICON [21], as an NSI-SOD method, performs poorly in many challenging scenarios, underscoring that directly applying NSI-SOD methods to ORSI images is a suboptimal solution. The visual results clearly validate the effectiveness of the proposed ORSIDiff, demonstrating its ability to handle the unique complexities of remote sensing imagery and outperform existing methods in diverse and challenging scenarios.

3) Confidence-Level Setting: Fig. 9 illustrates the gradual evolution of the prediction results at different time steps. Initially, when the noise level is high, the model generates a coarse mask with high uncertainty, reflecting its initial prediction about the salient object regions. As the number of iterations in the sampling process increases, the model progressively focuses on salient object regions and refines the local details of the mask by capturing fine-grained features of the foreground. Ultimately, the model achieves accurate segmentation by modeling the subtle differences between salient objects and the background. This gradual refinement process benefits from the layer-by-layer iterative nature of the diffusion model, which effectively guides the model to optimize the object region.

In our experiments, we empirically set $T = 10$. As shown in Fig. 9, there are some false positives due to the inherent complexity of the scenes, where depth confusion or interference between salient objects and the background can occur. To address this, we adopt a looser threshold of 0.4 (instead of 0.5) for M_a in (14), aiming to capture as many salient objects as possible. This strategy effectively mitigates the interference caused by outliers generated during the reverse denoising

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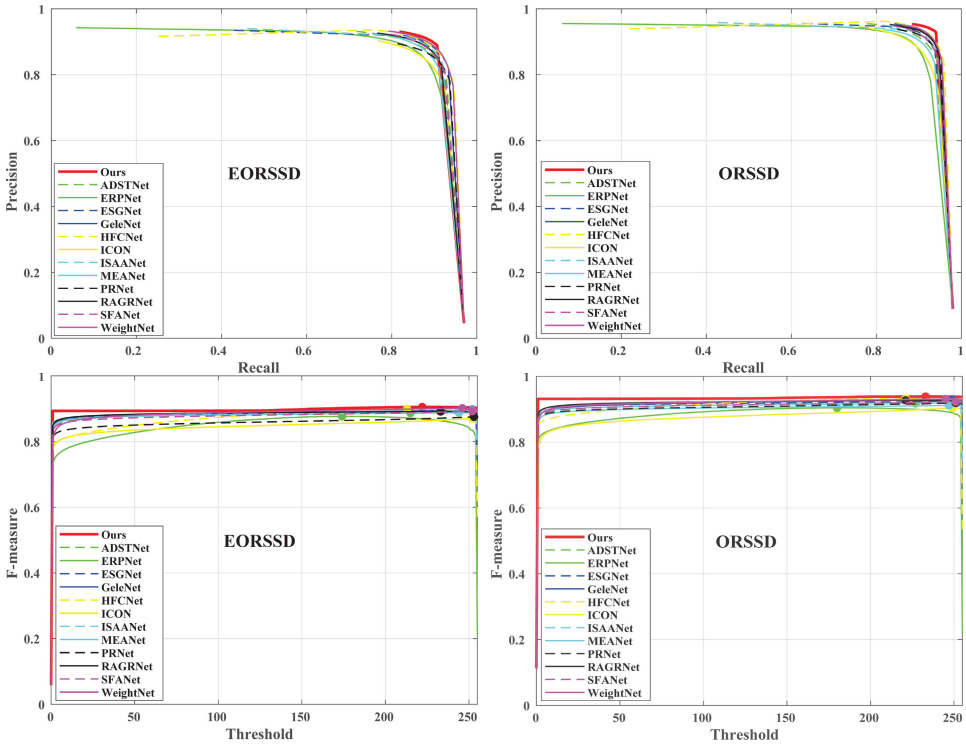


Fig. 7. P–R curves and F-measure curves of different methods. Our model (solid red line) achieves the highest precision and recall scores on the two benchmark datasets compared to other methods and consistently maintains the best performance as the F-measure threshold changes.

process, thereby enhancing the reliability and accuracy of the final detection results. This approach ensures that ORSIDiff maintains robust performance even in highly complex and challenging scenarios.

C. Ablation Studies

To comprehensively evaluate the contribution of each component and strategy in ORSIDiff, we conduct a series of ablation experiments by systematically removing or replacing individual components or strategies. To ensure a fair comparison, all ablation studies adhere to the same training strategy. The experiments are performed on the EORSSD dataset, a widely used benchmark for ORSI-SOD tasks. Through these experiments, we aim to quantify the impact of each element on the overall performance of ORSIDiff, providing insights into their respective roles and validating the effectiveness of our proposed framework.

1) Effectiveness of Key Components: ORSIDiff incorporates two critical components: the time step prompt (TSP) and the CAS. To evaluate the contribution and impact of each component, we conduct ablation experiments by removing TSP or CAS individually. As shown in Table II, the performance of ORSIDiff declines when TSP is removed (ID: 2). This is because the reverse denoising process of the DM is a continuous process dependent on time steps T. TSP plays a crucial role in helping the network distinguish between denoising processes at different time steps, enabling the generation of richer predictions and effectively addressing the pixel-level probability dependence problem. The results highlight the importance of TSP in maintaining the model's performance and robustness.

TABLE II
ABLATION EXPERIMENT OF TSP AND CAS

ID	CAS	TSP	EORSSD			
			$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$
1	✓	✓	0.9850	0.9005	0.9482	0.0042
2	✓		0.9814	0.8957	0.9447	0.0046
3		✓	0.9849	0.8997	0.9465	0.0050

After removing the CAS (ID: 3), our model exhibits a slight decline in the metrics $E_\xi^{\max}$, $F_\beta^{\max}$, and S_α, along with a significant drop in the metric $\mathcal{M}$. This is because CAS integrates diverse prediction results from a filtering perspective and optimizes the final predictions at the pixel level, thereby enhancing the network's overall performance. Importantly, CAS achieves these improvements without incurring additional computational costs. These results underscore the critical role of CAS in refining predictions and improving the reliability of the model, demonstrating the value of each component in driving performance gains.

2) Backbone Networks: To ensure a fair comparison, we replace the backbone of several SOTA methods, such as ISSANet, GeleNet, and ACCoNet, with the backbone STM used in ORSIDiff. These modified versions are named ISSANet (SMT), GeleNet (SMT), and ACCoNet (SMT), respectively. The results, as shown in Table III, demonstrate that ORSIDiff consistently achieves the best performance across all evaluation metrics. This validates the effectiveness of ORSIDiff and highlights the advantages of its design, including the integration of the SMT backbone and the novel components tailored for ORSI-SOD tasks.

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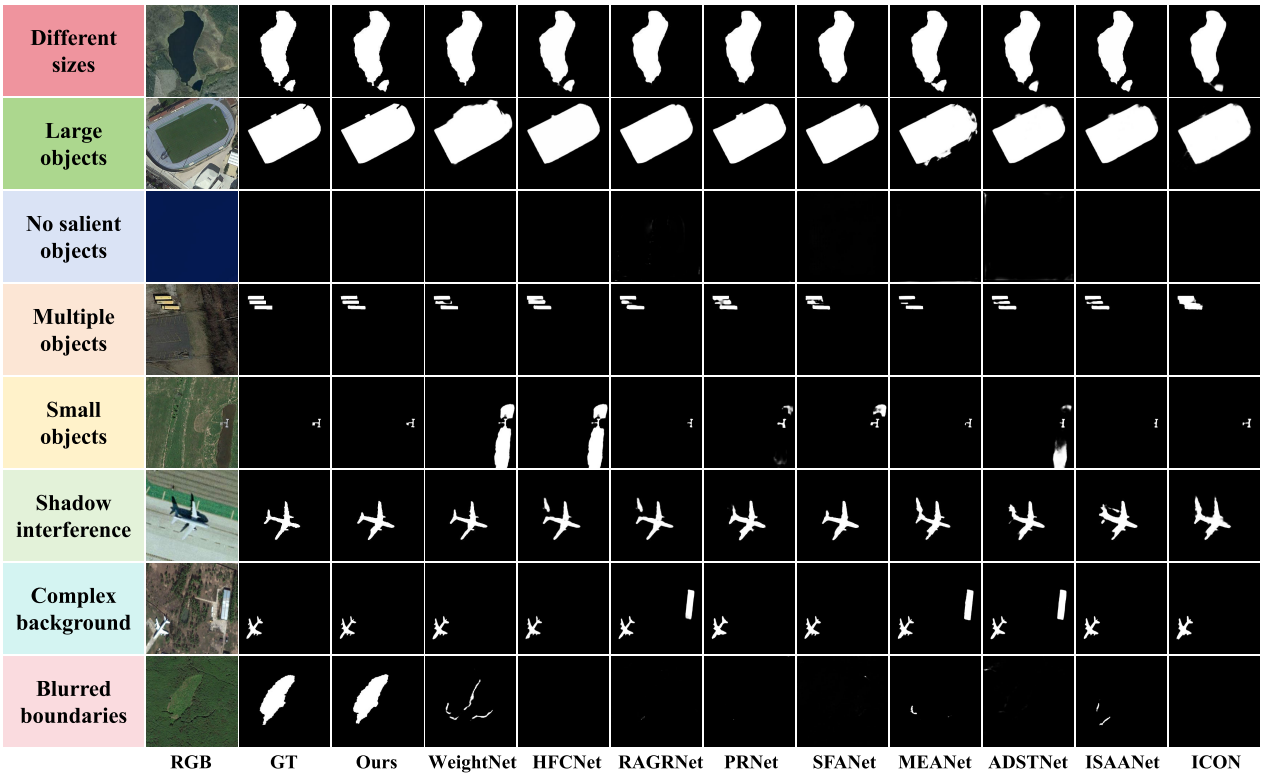


Fig. 8. Visualization of sample detection results across eight challenging scenarios, showcasing the robustness and versatility of ORSIDiff. These scenarios include: different sizes, large objects, no salient objects, multiple objects, small objects, shadow interference, complex background, and blurred boundaries.

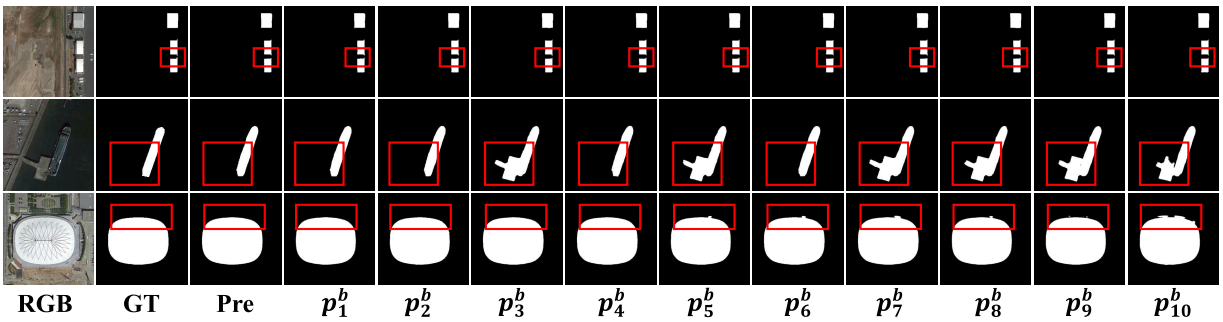


Fig. 9. Prediction results of the model at different time steps t. For the first row, the positive result is p_3^b, p_4^b, p_5^b, p_6^b, and p_{10}^b; for the second row, the positive result is p_1^b, p_2^b, p_4^b, and p_6^b; and for the third row, the positive result is p_1^b, p_2^b, p_3^b, and p_4^b. From these results, we observe that the final prediction $\hat{x}_0$ optimizes the T potential results p_t^b $(t = 1, \ldots, T)$. Here, we apply the CAS to filter out noise to a certain extent, ensuring that the final prediction is more reliable and accurate.

TABLE III

ABLATION EXPERIMENTS ON THE SAME BACKBONE, i.e., SMT. THE BEST RESULTS ARE SHOWN IN RED

Method	EORSSD							
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
GeleNet	0.9175	0.8627	0.8452	0.8238	0.9684	0.9631	0.9573	0.0075
ACCoNet	0.9251	0.8670	0.8470	0.8097	0.9795	0.9713	0.9532	0.0067
ISSANet	0.9317	0.8738	0.8569	0.8429	0.9742	0.9659	0.9558	0.0064
ORSIDiff	0.9482	0.9005	0.8885	0.8865	0.9850	0.9812	0.9831	0.0042

3) Effectiveness of Modules in Denoising Network: To evaluate the contribution and impact of each component in the denoising network, we conduct a series of ablation studies on the EORSSD dataset. Specifically, we start with a baseline model (ID: 1), which consists of a backbone and ordinary CLs, and gradually extend it by incorporating different components, including the FGEM, RM, and PRM.

As shown in Table IV, adding FGEM (ID: 2) significantly improves the performance of the baseline. Specifically, for ID 2, the metrics $E_\xi^{\max}$, $F_\beta^{\max}$, S_α, and $\mathcal{M}$ improve by 2.19%, 8.9%, 3.9%, and 0.36%, respectively. This improvement is attributed to FGEM's ability to explore intrinsic consistency information within features and model contextual information at different scales. By enhancing the dominant information in features, FGEM effectively boosts detection performance, demonstrating its critical role in the network's overall effectiveness.

For ID 3, where the RM is added to ID 2, the metrics $E_\xi^{\max}$, $F_\beta^{\max}$, S_α, and $\mathcal{M}$ improve by 3.69%, 14.71%, 8.50%, and 0.63%, respectively. This significant improvement highlights

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TABLE IV

ABLATION EXPERIMENT OF DIFFERENT MODULES, INCLUDING THE FGEM, RM, AND PRM. THE BEST RESULTS ARE SHOWN IN RED

ID	Baseline	FGEM	RM	PRM	EORSSD $E_\xi^{max}\uparrow$	EORSSD $F_\beta^{max}\uparrow$	EORSSD $S_\alpha\uparrow$	EORSSD $\mathcal{M}\downarrow$
1	✓				0.9219	0.6550	0.8177	0.0146
2	✓	✓			0.9438	0.7440	0.8567	0.0110
3	✓	✓	✓		0.9807	0.8911	0.9417	0.0047
4	✓	✓	✓	✓	0.9850	0.9005	0.9482	0.0042

TABLE V

ABLATION EXPERIMENTS OF THE ENCODER AND DECODER OF DENOISING NETWORK. THE BEST RESULT IS SHOWN IN RED

ID	Components	EORSSD $E_\xi^{max}\uparrow$	EORSSD $F_\beta^{max}\uparrow$	EORSSD $S_\alpha\uparrow$	EORSSD $\mathcal{M}\downarrow$
1	PVTv2-b4	0.9856	0.8997	0.9452	0.0044
2	CNN layer	0.9727	0.8683	0.9295	0.0076
3	ORSIDiff	0.9850	0.9005	0.9482	0.0042

TABLE VI

ABLATION EXPERIMENT OF INPUT IMAGE SIZE. THE BEST RESULTS ARE SHOWN IN RED

ID	Image Size	EORSSD $E_\xi^{max}\uparrow$	EORSSD $F_\beta^{max}\uparrow$	EORSSD $S_\alpha\uparrow$	EORSSD $\mathcal{M}\downarrow$
1	256	0.9825	0.8973	0.9455	0.0045
2	352	0.9844	0.8990	0.9469	0.0043
3	ORSIDiff(384)	0.9850	0.9005	0.9482	0.0042

the effectiveness of RM in leveraging noisy mask features to guide the decoding process, enabling precise local modeling and refinement of feature details. These results confirm that RM plays a crucial role in enhancing model performance, particularly in capturing and refining fine-grained details essential for accurate SOD.

For ID 4, where the PRM is added to ID 3, the metrics $E_\xi^{\max}$, $F_\beta^{\max}$, S_α, and $\mathcal{M}$ improve by 0.43%, 0.94%, 0.65%, and 0.05%, respectively. The strength of PRM lies in its use of the RM combined with attention mechanisms to assist in global perception modeling. This enhances the overall salient information while suppressing background noise under the guidance of a global perspective. Additionally, PRM leverages the rich detailed information typically present in the lowest-level features, capturing as much detail as possible from a global standpoint. These results demonstrate that PRM contributes to the model's performance, further refining the detection accuracy and robustness.

The previous analysis highlights the individual contributions of each component—FGEM, RM, and PRM—to the model's performance. When these components work collaboratively, they synergistically enhance the model's ability to detect salient objects in complex remote sensing scenes. This integrated approach significantly improves overall performance, demonstrating the effectiveness of the proposed architecture in addressing the unique challenges of ORSI-SOD tasks.

4) Effectiveness of the Encoder and Decoder of Denoising Network: To further evaluate the effectiveness of the encoder and decoder from the denoising network, we conduct the following experiments: (ID: 1) replace the SMT backbone with PVTv2 as the encoder; (ID: 2) replace the proposed decoder with a traditional decoder composed of standard CLs; and (ID: 3) the proposed ORSIDiff with the full architecture, including the SMT backbone and the novel decoder design. All ablation experiments are compared against (ID: 3) to assess the impact of each component. This comparison allows us to quantify the contributions of the SMT backbone and the proposed decoder design to the overall performance of ORSIDiff, providing insights into their respective roles in enhancing SOD in remote sensing imagery.

The experimental results, as shown in Table V, reveal the following insights. (ID: 1) demonstrates the robustness of the denoising network when using different encoders, confirming that the network can adapt to various backbone architectures while maintaining competitive performance. (ID: 2) provides a baseline result, where the decoder is composed of standard CLs without the sophisticated modules used in traditional diffusion models. The performance gap between (ID: 2) and (ID: 3) highlights the effectiveness of the decoder and the FGEM designed in the denoising network. These components significantly improve the model's ability to refine predictions and capture salient object details.

Overall, these results underscore the importance of the proposed decoder and FGEM in enhancing the performance of ORSIDiff, validating their contributions to the network's effectiveness.

5) Image Size: Since different SOTA methods may use varying image sizes, we conduct experiments to evaluate the performance of ORSIDiff under different image resolutions. As shown in Table VI, we test three scenarios: (ID: 1) image size 256 × 256; (ID: 2) image size 352 × 352; and (ID: 3) image size 384 × 384. From the results in Table VI, it is evident that changes in image size have minimal impact on the performance of ORSIDiff. This demonstrates the robustness of ORSIDiff, as it consistently delivers high performance across different input resolutions. This adaptability makes ORSIDiff a reliable and versatile solution for ORSI-SOD tasks, regardless of the input image dimensions.

6) Time Steps: To verify the rationality of the selected diffusion step $T = 10$, we construct a comparative experiment to explore the impact of the number of diffusion steps on model performance. Table VII presents the results of three groups of control experiments: 1) $T = 5$; 2) $T = 15$; and 3) $T = 10$.

The experimental results indicate that the indicators of (ID: 3) are the best compared to those of the other two cases. This phenomenon can be explained from the perspective of diffusion dynamics. When the number of diffusion steps is small ($T = 5$), the noise removal process becomes overly hasty. As a result, the model cannot adequately capture the probability distribution differences between the salient target and the background within such a limited number of steps. This leads to insufficient feature separation, particularly in complex scenes where highly unstructured backgrounds and multiscale targets coexist. On the contrary, when the number

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TABLE VII

ABLATION EXPERIMENT OF THE NUMBER OF TIME STEPS. THE BEST RESULTS ARE SHOWN IN RED

ID	Time Steps	FPS (GPU)	EORSSD			
			$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$
1	5	12	0.9846	0.8976	0.9463	0.0043
2	15	3	0.9839	0.8962	0.9459	0.0040
3	ORSIDiff(10)	8	0.9850	0.9005	0.9482	0.0042

TABLE VIII

ABLATION EXPERIMENT OF LOSS FUNCTION. THE BEST RESULTS ARE SHOWN IN RED

ID	Variants	EORSSD			
		$E_\xi^{max}\uparrow$	$F_\beta^{max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$
1	IoU	0.9819	0.8914	0.9427	0.0045
2	BCE	0.9824	0.8966	0.9463	0.0043
3	BCE+IoU	0.9850	0.9005	0.9482	0.0042

of diffusion steps is too large ($T = 15$), although more detailed noise removal is theoretically achievable, two adverse effects arise in practice. First, excessive noise reduction results in the loss of high-frequency detailed information, blurring target boundaries, and reducing contrast. Second, due to the Markov chain nature of the diffusion model, longer reverse inference paths accumulate more prediction errors. These errors are amplified and propagated at each time step, ultimately leading to issues such as significant positional offsets in target areas and boundary instability. Therefore, in our experiments, we set the time step to $T = 10$ as an optimal balance between noise removal and preserving critical details.

7) Loss Function: To verify the rationality of the loss function used in ORSIDiff, we conduct experiments to evaluate its performance under different loss functions. As shown in Table VIII, we test three scenarios: (ID: 1) using only IoU loss; (ID: 2) using only binary cross-entropy loss (BCE) loss; and (ID: 3) using IoU loss and BCE loss. From the results in Table VIII, it is clear that both loss functions contribute to improving the performance of ORSIDiff. Specifically, BCE loss focuses on the classification accuracy of each pixel, ensuring precise pixel-level predictions; IoU loss emphasizes matching at the object level, enhancing the model's ability to accurately localize and segment salient objects. The combination of BCE and IoU losses allows the model to comprehensively consider both pixel-level and object-level information, enabling it to learn the characteristics of salient objects more thoroughly. This hybrid approach results in superior performance, demonstrating the effectiveness of the combined loss function in ORSIDiff.

V. CONCLUSION

This article introduces a novel paradigm that reformulates the ORSI-SOD task as a mask generation problem based on the diffusion model. We propose an innovative diffusion model, named ORSIDiff, marking the first application of diffusion models to the ORSI-SOD task. To address the limitations of traditional diffusion models in SOD, we design a denoising network that integrates ORSI into the denoising process, enhancing the model's discriminative capabilities. This network employs a hybrid backbone architecture, combining strong global and local modeling abilities to better capture both the detailed features and overall attributes of salient objects. Additionally, it comprehensively understands the differences between salient objects and complex backgrounds, improving detection accuracy. Furthermore, we introduce a CAS to mitigate the interference caused by outliers generated at different steps during the reverse denoising process. This strategy enhances the reliability and accuracy of the detection results by filtering out inconsistent predictions. Thanks to the synergistic collaboration of its components, ORSIDiff effectively addresses the multiple challenges inherent in ORSI-SOD, establishing itself as a robust and successful detector for this task. The extensive experimental results demonstrate the superior performance of ORSIDiff and validate the effectiveness of each component, highlighting its potential for advancing ORSI-SOD research and applications.

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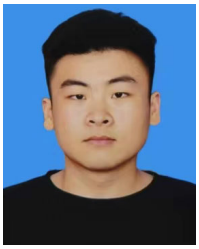
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Jinyu Han received the B.S. degree from Luoyang Normal University, Luoyang, China, in 2022. He is currently pursuing the M.S. degree with the School of Information and Communication Engineering, Dalian Minzu University, Dalian, China.

His research interests include pattern recognition and deep learning.



Jing Sun received the Ph.D. degree from the DUT-RU International School of Information Science and Engineering, Dalian University of Technology, Dalian, China, in 2021.

She is currently a Lecturer with the School of Information and Communication Engineering, Dalian Minzu University, Dalian. Her research interests include pattern recognition and transfer learning. She has authored or co-authored over ten journal articles and conference papers in these areas, including IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, *Knowledge-Based Systems*, and ACM Multimedia.



Fasheng Wang received the Ph.D. degree in computer science from Dalian Maritime University, Dalian, China, in 2015.

From August 2007 to June 2017, he was with the Department of Computer Science and Technology, Dalian Neusoft University of Information, Dalian. He is currently a Professor with the School of Information and Communication Engineering, Dalian Minzu University, Dalian. His research interests mainly focus on computer vision and pattern recognition.



Fuming Sun received the Ph.D. degree from the University of Science and Technology of China, Hefei, China, in 2007.

From 2007 to 2018, he worked at Liaoning University of Technology, Jinzhou, China. He is currently a Professor with the School of Information and Communication Engineering, Dalian Minzu University, Dalian, China. His research interests include image content analysis and pattern recognition. He has authored or co-authored over 70 journals and conference papers in these areas, including IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON IMAGE PROCESSING, *International Journal of Computer Vision*, and ACM Multimedia.



Haojie Li received the Ph.D. degree from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2007.

From 2007 to 2009, he was a Research Fellow with the School of Computing, National University of Singapore, Singapore. From 2010 to 2023, he was with the DUT-RU International School of Information Science and Engineering, Dalian University of Technology, Dalian, China. He is currently a Professor with the School of Computer Science and Engineering, Shandong University of Science and Technology, Qingdao, China. His research interests include social media computing and computer vision.

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